

EX. NO. 08 - STEGEXPOSE STEGANALYSIS

Experiment Name	Use StegExpose to Detect Hidden Data in Images
Aim	To use StegExpose to analyze digital images and detect possible hidden data using statistical steganalysis.
Tool Used	Java Runtime Environment (JRE), StegExpose, Windows Command Prompt, Test images

OBJECTIVE

- To examine digital images for possible steganographic content.
- To analyze the statistical properties of the given images using StegExpose.
- To obtain the Fusion score for each image.
- To compare the obtained scores with the default steganographic detection threshold.

PROCEDURE

Step 1: Verify Java Installation

Java installation was verified using the following command:

```
java -version
```

The installed Java environment was displayed successfully, confirming that Java was available for executing the StegExpose JAR file.

Screenshot 1 - Java version verification

Step 2: Create the Experiment Directory

A separate directory was created on the D: drive for Experiment 8.

```
D:
mkdir DF_Experiment_8
cd DF_Experiment_8
```

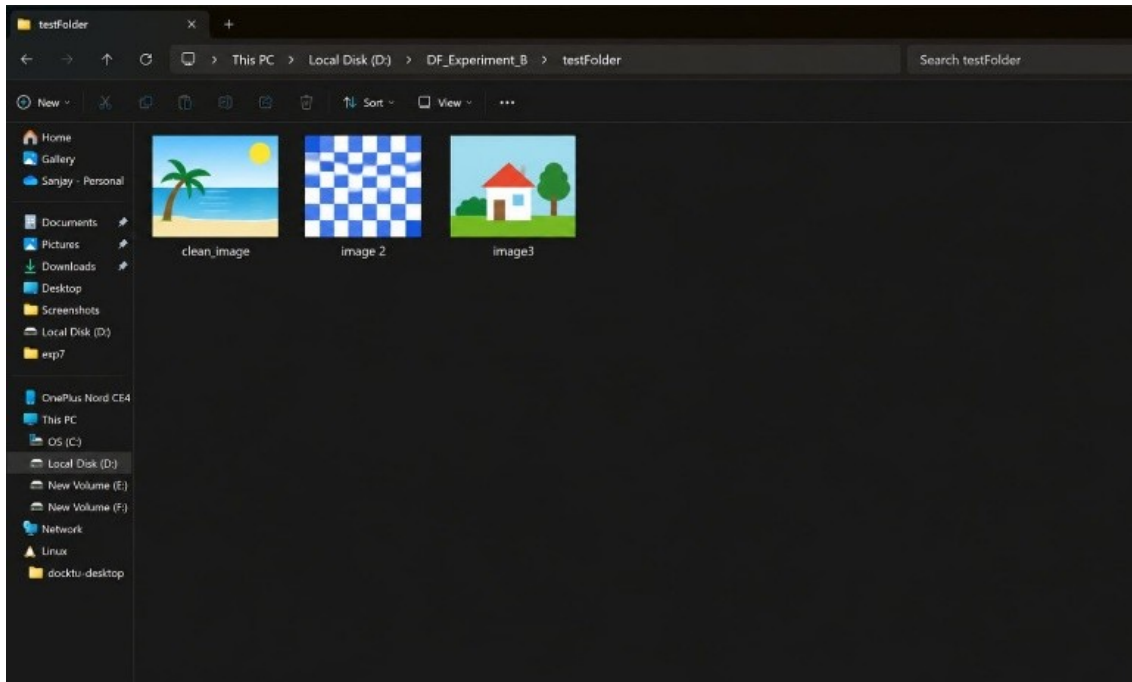
The experiment directory used was:

```
D:\DF_Experiment_8
```

A subdirectory named testFolder was created to store the test images.

```
mkdir testFolder
```

The StegExpose JAR file and test images were placed in the experiment directory and testFolder, respectively.



Screenshot 2 - Experiment folder and test images

Step 3: Execute StegExpose

The StegExpose program was executed on the test image folder using:

```
java -jar StegExpose.jar testFolder
```

The command completed successfully without displaying an error.

```
D:\DF_Experiment_8>D:
D:\DF_Experiment_8>cd \DF_Experiment_8
D:\DF_Experiment_8>java -jar StegExpose.jar testFolder
D:\DF_Experiment_8>java -jar StegExpose.jar testFolder default default results.csv
```

Screenshot 3 - StegExpose execution

Step 4: Generate the Steganalysis Results

To obtain detailed statistical results in CSV format, the following command was executed:

```
java -jar StegExpose.jar testFolder default default results.csv
```

The command generated the file:

```
results.csv
```

The generated CSV file contained the statistical analysis values and Fusion scores.

The default StegExpose threshold used in the experiment was 0.2. All three images produced Fusion scores below 0.2, and the Above stego threshold? field was reported as false for all three images.

```
D:\DF_Experiment_8>type results.csv
File name,Above stego threshold?,Secret message size in bytes (ignore for clean files),Primary Sets,Chi Square,Sample Pairs,RS analysis,Fusion (mean)
clean_image.png,false,2,0.001355604875670536,0.0,0.0010629026672680287,0.00146195733339978242,9.701162192340973E-4
image 2.png,false,0,0.0,0.0,0.0,0.0,0.0
images 3.png,false,10,0.00750732396181886,0.0,0.009029365936654453,0.008589190195225856,0.006281470023424792
D:\DF_Experiment_8>
```

Screenshot 4 - results.csv output and Fusion scores

File name	Above stego threshold?	Primary Set	Chi Square	Sample Po	RS analysis	Fusion (mean)
clean_image	FALSE	2	0.001350	0	0.001065	0.001462
image 2.pr	FALSE	0	0	0	0	0
image 3.pr	FALSE	10	0.007507	0	0.009029	0.008589

Screenshot 5 - Results displayed in spreadsheet form

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

StegExpose was successfully used to perform statistical steganalysis on the three test images. The generated results.csv file provided the statistical analysis values and Fusion scores. All three images were reported as False under Above stego threshold?, indicating that none of the analyzed images exceeded the configured StegExpose threshold of 0.2.